

Solving Linear Equations

T.T.

A system is interpreted by rows and columns. Elimination produces echelon form, pivots determine existence and uniqueness, and matrix factorizations organize the same computation.

SYSTEMS AND ELIMINATION

$$Ax = b, \quad Ax = x_1 a_1 + \cdots + x_n a_n.$$

Row operations:

interchange rows; multiply a row by a nonzero constant; add a multiple of one row to another.

Rank criterion:

$Ax = b$ is solvable iff $\text{rank}(A) = \text{rank}([A \ b])$.

Pivot theorem:

free variables occur in non-pivot columns; if r is the rank, the solution set has $n - r$ free parameters.

$$Ax = 0 \Rightarrow x = x_{\text{pivot}} + x_{\text{free}}; \quad x = x_p + x_n, \quad Ax_n = 0.$$

Rouché–Capelli conclusion:

a consistent system has either one solution (no free variables) or infinitely many solutions.

MATRIX OPERATIONS

$$A^{-1}A = I, \quad (AB)^{-1} = B^{-1}A^{-1}, \quad (A^T)^{-1} = (A^{-1})^T.$$

$$E_{ij}^{-1} = E_{ij}(-c), \quad E_i(c)^{-1} = E_i(c^{-1}), \quad P^{-1} = P^T.$$

Elimination factorization:

$EA = U$, hence $A = LU$; with row exchanges, $PA = LU$.

LU solution:

solve $Ly = b$ followed by $Ux = y$.

SPECIAL MATRICES

$$A^T A \text{ is symmetric, } A = S + K, \quad S = \frac{A + A^T}{2}, \quad K = \frac{A - A^T}{2}.$$

Orthogonal matrix:

$Q^T Q = I \iff Q^{-1} = Q^T$; it preserves dot products and lengths.

$$\det(AB) = \det A \det B, \quad A^{-1} = \frac{1}{\det A} \text{Cof}(A)^T.$$